PS2

Q 1.1

(a) $V(a,z) = \max_{a' \in A(a,z)} \{ u(z + Ra - a') + \beta E[V(a', z') \mid z] \}$

(a,z) is the state variable because earlier history provides no information. Hence consumption and income is determined by (a,z)

(b) Take first derivative w.r.t a' :

$$-u'(z + Ra - a') + \beta \cdot E\left[\frac{\partial V(a',z')}{\partial a'} \mid z\right] = 0$$

(c) $$\frac{\partial V(a,z)}{\partial a} = u'(c) \cdot \left(R - \frac{da'}{da}\right) + \beta \cdot E\left[\frac{dV}{da} \mid z\right] \cdot \frac{da'}{da}$$

by envelope theorem, ~~$\frac{da'}{da} = 0$~~ ~~, we have :~~

$$V_a'(a,z) = R\,u'(c) + \frac{da'}{da}\underbrace{\left[-u'(c) + \beta E\left[\frac{\partial V(a',z')}{\partial a'} \mid z\right]\right]}_{=0 \text{ by (b)}} = R\,u'(c)$$

(d) Substitute (c) into (b) :

$$u'(c) = R\beta E[u'(c') \mid z]$$

Q 1.2

(a) The Euler equation is: $e^{-rc} = \beta R E[e^{-rc'} \mid x, e]$

Since $a' = x - c = (1-\mu)x - ve - f$, we have

$x' = \bar{z} + e' + Ra'$, $e' = \rho e + \sigma_\varepsilon \varepsilon'$

Hence $c' = \mu x' + ve' + f = \mu R(1-\mu)x + [\mu(\rho - Rv) + v\rho]e +$

$\mu(\bar{z} - Rf) + f + (\mu + v)\sigma_\varepsilon \varepsilon'$

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(b) We have $E[e^{rc'} | x, e] = \exp\left(-rm' + \frac{r^2}{2}(\mu+v)^2\sigma_\varepsilon^2\right)$

Take log: $\mu x + ve + f = m' - \frac{\ln(\beta R)}{r} - \frac{r}{2}(\mu+v)^2\sigma_\varepsilon^2$

$$\Rightarrow \begin{cases} \mu = \mu R(1-\mu) & \Rightarrow \mu = \frac{r}{R} \\ v = \mu(\rho - Rv) + v\rho & \Rightarrow v = \frac{\rho\mu}{R-\rho} \\ f = \mu(\bar{z} - Rf) + f - \frac{\ln(\beta R)}{r} - \frac{r}{2}(\mu+v)^2\sigma_\varepsilon^2 & \Rightarrow f = \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{rr} - \frac{r(\mu+v)^2\sigma_\varepsilon^2}{2r} \end{cases}$$

(c) μ is the marginal propensity to consume. So given one unit of additional cash, households would consume $\frac{r}{R}$ and save $\frac{1}{R}$.

(d) $\sigma_\varepsilon \varepsilon$ raises consumption through e and x in the sense that

if $\Delta x = \Delta e = \sigma_\varepsilon \varepsilon$, then $\Delta c = (\mu+v)\Delta e$,

$$\mu + v = \frac{r}{R}\left(1 + \frac{\rho}{R-\rho}\right) = \frac{r}{R-\rho} \quad \Rightarrow \quad \frac{\partial(\mu+v)}{\partial\rho} = \frac{r}{(R-\rho)^2} > 0$$

Consumption responds strongly as more persistent income news raises expected future resources by more.

As $\rho \to 0$, the shock is transitory and the response goes to $\frac{1}{R}$

As $\rho \to 1^-$, the response $\mu+v$ goes to 1.

(e) Let $\rho = 0$, then $v = 0$. $\sigma_\varepsilon = 0$, the policy is thus:

$$c(x) = \frac{r}{R}x + \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{rr} - \frac{r\sigma_\varepsilon^2}{2R^2}$$

$$a' = x - c(x)$$

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2.1 Exact policy of the seven-state Rouwenhorst chain

Here $\gamma = 2$, $r = 0.04$, $R = 1.04$, $\beta = 1/R$, $\bar{z} = 1$, $\sigma = 0.1$, and $\rho = 0.9$; hence $\sigma_\varepsilon = \sigma\sqrt{1-\rho^2} \simeq 0.0435890$. The continuous-normal formulas in Part 1 give

$$(\mu, \nu, f)_{\text{normal}} = (0.0384615385,\ 0.2472527473,\ 0.9576609105). \tag{1}$$

The Rouwenhorst chain has $N_z = 7$ states $e_j = (2j - n)\delta$, $j = 0, \ldots, n$, with $n = N_z - 1 = 6$, $\delta = \sigma/\sqrt{n}$, and persistence parameter $p = (1 + \rho)/2$. To account for its *discrete* conditional moment-generating function exactly, set

$$H_+(t) = pe^{t\delta} + (1-p)e^{-t\delta}, \qquad H_-(t) = (1-p)e^{t\delta} + pe^{-t\delta}.$$

For a state containing j positive and $n - j$ negative components, the Rouwenhorst transition gives

$$\begin{aligned} \mathbb{E}[e^{te'} \mid e_j] &= H_+(t)^j H_-(t)^{n-j} = \exp(A(t) + B(t)e_j), \\ A(t) &= \frac{n}{2}\ln(H_+(t)H_-(t)), \qquad B(t) = \frac{1}{2\delta}\ln\frac{H_+(t)}{H_-(t)}. \end{aligned} \tag{2}$$

Using $c' = \mu R(1-\mu)x - \mu R\nu e + \mu(\bar{z} - Rf) + f + (\mu + \nu)e'$ in the Euler equation, matching coefficients again implies

$$\boxed{\mu = \frac{r}{R}, \qquad \nu_{N_z} = \frac{B(\gamma(\mu + \nu_{N_z}))}{\gamma R}, \qquad f_{N_z} = \frac{\bar{z}}{R} - \frac{\ln(\beta R)}{\gamma r} - \frac{A(-\gamma(\mu + \nu_{N_z}))}{\gamma r}.} \tag{3}$$

The resulting values and differences (*discrete minus continuous*) are

Coefficient	Continuous normal	Rouwenhorst ($N_z = 7$)	Difference
μ	0.0384615385	0.0384615385	0
ν	0.2472527473	0.2471894794	-6.3268×10^{-5}
f	0.9576609105	0.9576621251	$+1.2146 \times 10^{-6}$
$\mu + \nu$	0.2857142857	0.2856510178	-6.3268×10^{-5}

At fixed (x, e) the consumption-policy difference is $\Delta c = (\nu_{N_z} - \nu)e + (f_{N_z} - f)$. Since $|e| \le \sigma\sqrt{6} \simeq 0.244949$ on the seven-state chain,

$$\max_{e \text{ on chain}} |\Delta c| \simeq 1.67 \times 10^{-5}.$$

This is extremely small compared with typical consumption of order one; the discretization difference is not economically meaningful at the stated calibration. (The notebook may calculate these coefficients directly from the transition matrix; (3) is the equivalent analytic calculation.)

3.1 From CARA to CRRA

The new model sets $\gamma = 2$, $r = 0.03$, $R = 1.03$, $\beta = 0.96$, and $\ln z' = \rho \ln z + \sigma_\varepsilon \varepsilon'$ with $\rho = 0.9$ and $\sigma_\varepsilon = 0.2$.

(a) Utility, marginal utility, and inverse marginal utility.

$$\begin{aligned} &\text{CARA:} \quad u(c) = -\frac{e^{-\gamma c}}{\gamma}, \qquad u_c(c) = e^{-\gamma c}, \qquad u_c^{-1}(y) = -\frac{\ln y}{\gamma}. \\ &\qquad\qquad c \in \mathbb{R}, \quad y > 0. \\ &\text{CRRA:} \quad u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \qquad u_c(c) = c^{-\gamma}, \qquad u_c^{-1}(y) = y^{-1/\gamma}. \\ &\qquad\qquad c > 0, \quad y > 0 \quad (\gamma = 2:\ u(c) = -1/c,\ u_c(c) = c^{-2}). \end{aligned}$$

CARA permits negative consumption mathematically; CRRA does not, and utility tends to $-\infty$ as $c \downarrow 0$ for the assigned $\gamma = 2$.

(b) Income process and discretization. CARA approximates the level deviation $e = z - \bar{z}$, following $e' = \rho e + \sigma_\varepsilon \varepsilon'$, by a seven-point Rouwenhorst chain; income in levels is $z_m = \bar{z} + e_m$. CRRA instead approximates the *log-income* state $\ell = \ln z$, following $\ell' = \rho \ell + \sigma_\varepsilon \varepsilon'$, by a seven-point chain. Income at state m is $z_m = e^{\ell_m} > 0$ (not ℓ_m or $1 + \ell_m$). The CRRA model does not have normally distributed *level* income.

(c) Asset grid. The CARA household has no economic borrowing constraint ($a' \in \mathbb{R}$); Part 2 uses a computationally truncated, *uniform* grid $a_i \in [-15, 25]$ with $N_a = 400$. For CRRA, the imposed borrowing limit fixes the lower endpoint at $a_{\min} = 0$. There is no finite theoretical upper asset bound, so $a_{\max}$ is a numerical truncation chosen large enough that the upper endpoint does not distort the relevant policies/diagnostics; its exact numerical value must come from the notebook setup rather than the problem statement. A *nonuniform* grid should concentrate points near zero, where the borrowing constraint and precautionary-saving behavior make policy functions most curved.

(d) Constraint and feasible consumption. In either model $a' = z_m + Ra_i - c$ and $x_{im} = z_m + Ra_i$. With no CARA borrowing limit and $c \in \mathbb{R}$, all real a' are theoretically admissible. On the CARA numerical choice interval $[-15, 25]$, consumption would be restricted to $x_{im} - 25 \leq c \leq x_{im} + 15$ by that numerical interval alone.

In CRRA, $a' \geq 0$ and $c > 0$ imply

$$\boxed{0 < c \leq x_{im}, \qquad a' = x_{im} - c \geq 0.}$$

If an algorithm also restricts next assets to the computational grid $a' \leq a_{\max}$, the full condition is

$$\boxed{a' \in [0, a_{\max}], \quad c = x_{im} - a' > 0, \quad x_{im} - a_{\max} \leq c \leq x_{im}.}$$

Thus for $x_{im} \leq a_{\max}$, the feasible consumption interval is $(0, x_{im}]$; for $x_{im} > a_{\max}$, it is $[x_{im} - a_{\max}, x_{im}]$. In particular, at a binding borrowing constraint ($a' = 0$), $c = x_{im}$, while zero or negative consumption is inadmissible.